

作业纸

课程名称: _____

班级: _____

教学班级: _____

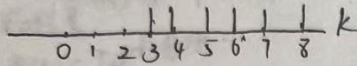
姓名: _____

学号: _____

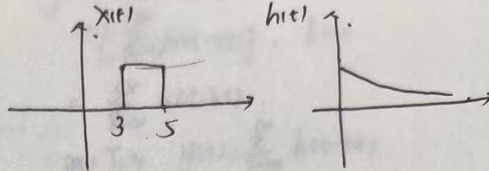
第 _____ 页

2-4

$$y[n] = x[n] * h[n]$$



2-11 $x(t) = u(t-3) - u(t-5)$, $h(t) = e^{-t} u(t)$.



(a) 计算 $y(t) = x(t) * h(t)$.

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

① $t \leq 3$, $y(t) = 0$

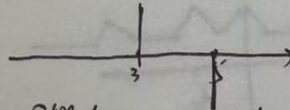
② $3 \leq t \leq 5$, $y(t) = \int_3^t 1 \cdot e^{-3(t-\tau)} d\tau$

$$y(t) = \frac{1}{3} e^{-3(t-3)} - \frac{1}{3} e^{-3(t-5)}$$

(b) $\left[\frac{d(x(t) * h(t))}{dt} \right]$

$$\frac{dx(t)}{dt} = \delta(t-3) - \delta(t-5) = h(t)$$

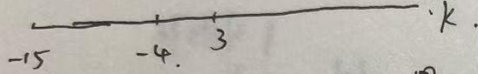
$h(t)$



$$\int_{-\infty}^{\infty} h(\tau) \delta(t-3-\tau) d\tau = h(t-3)$$

$$\therefore \frac{dy(t)}{dt} = h(t-3) - h(t-5) = e^{-3(t-3)} u(t-3) - e^{-3(t-5)} u(t-5)$$

(c) $g(t) = \frac{dy(t)}{dt}$



① $n < 7$ 无解

② $7 \leq n \leq 12$

$$y[n] = (n-6) \cdot 3$$

③ $12 < n \leq 18$

$$y[n] = 6$$

④ $18 < n \leq 23$

$$y[n] = (24-n) \cdot 3$$

⑤ $n > 23$

无解

$$\Rightarrow y[n] = \begin{cases} n-6, & 7 \leq n \leq 12 \\ 6, & 12 < n \leq 18 \\ 24-n, & 18 < n \leq 23 \\ 0, & \text{其他} \end{cases}$$

联系方式: _____

作业纸

课程名称: _____

班级: _____

教学班级: _____

姓名: _____

学号: _____

第 _____ 页

2.17 $\frac{dy(t)}{dt} + 4y(t) = x(t)$

2-23

(a) $x(t) = e^{(-1+3j)t} u(t)$

① 找特解, 即一个与输入形式相同
的信号.

$y_p(t) = \tilde{\gamma} e^{(-1+3j)t}$

$\Rightarrow \tilde{\gamma}(-1+3j)e^{(-1+3j)t} + 4\tilde{\gamma}e^{(-1+3j)t} = e^{(-1+3j)t}$

$3\tilde{\gamma} + 3j\tilde{\gamma} = 1$

$\therefore \tilde{\gamma} = \frac{1}{3(1+j)} = \frac{1-j}{6}$

$y_p(t) = \frac{1-j}{6} e^{(-1+3j)t}$

$y_h(t) = Ae^{st}$

$Ase^{st} + 4Ae^{st} = 0$

$\therefore s = -4$

$y_h(t) = Ae^{-4t}$

又 $y(0) = 0$

$\therefore A + \frac{1-j}{6} = 0 \quad A = -\frac{1-j}{6}$

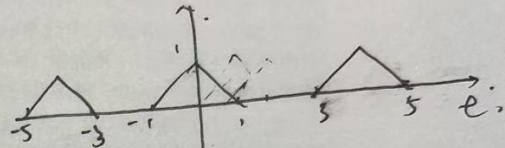
$\therefore y(t) = \frac{1-j}{6} [e^{(-1+3j)t} - e^{-4t}] u(t)$

$y(t) = x(t) * h(t)$

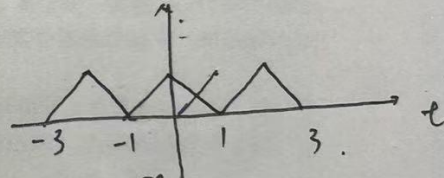
$= \left[\sum_{k=-\infty}^{\infty} \delta(t-kT) \right] * h(t)$

$= \sum_{k=-\infty}^{\infty} h(t-kT)$

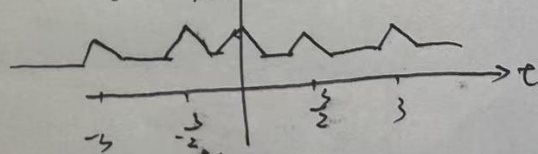
(a) $T=4, y(t) = \sum_{k=-\infty}^{\infty} h(t-4k)$



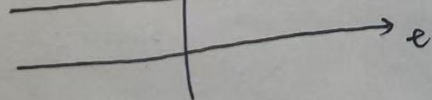
(b) $y(t) = \sum_{k=-\infty}^{\infty} h(t-2k)$



(c) $y(t) = \sum_{k=-\infty}^{\infty} h(t - \frac{5}{2}k)$



(d) $y(t) = \sum_{k=-\infty}^{\infty} h(t+k)$



联系方式: _____